

Utkarsh Singh

Software Engineer (Full-Stack, Python, TypeScript/React, AI/LLM Applications, Cloud-Native)

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Summary

Full-stack software engineer with **3+ years** of experience building, shipping, and operating scalable web applications and cloud-native services. Proficient in **Python** and **TypeScript/React** across the stack – from designing RESTful APIs and data-driven backends to building responsive, component-based UIs. Experienced in integrating **AI/LLM capabilities** into production applications and working directly with customers to iterate in live environments. Strong foundation in data structures, algorithms, design patterns, and object-oriented architecture. Proven track record of owning features end-to-end – from design through delivery – with automated testing, CI/CD, and production-grade observability baked in. **MSc Cloud Computing; AWS Solutions Architect – Professional**. Dublin-based (Stamp 1G) – eligible for full-time employment in Ireland.

Technical Skills

- **Languages & Frameworks:** Python (FastAPI, Flask, Django), TypeScript/JavaScript (React 19, React Native, Node.js/Express), Java, SQL, Bash
- **Front-End:** React, React Native, Tailwind CSS, Zustand/Redux, Framer Motion; responsive design, component architecture, state management, A/B testing
- **Databases & Data:** PostgreSQL, MySQL, DynamoDB, Redis, Firestore, Supabase; schema design, query optimization, ETL, PySpark, data pipelines
- **AI & ML:** LLM integration (multi-model consensus, prompt engineering), ONNX inference, Gemini API, scikit-learn, TensorFlow/Keras, NLP, RAG pipelines
- **Testing:** Jest (3,000+ tests), Maestro (E2E), pytest, Playwright, JUnit; SAST/DAST (SonarCloud, OWASP ZAP); TDD, code review
- **CI/CD & DevOps:** GitHub Actions, Jenkins, AWS CodePipeline, EAS Build; automated test gates, blue/green and canary deployments
- **Cloud:** AWS (EC2, ECS/Fargate, EKS, Lambda, RDS, S3, API Gateway, SQS/SNS, CloudFront, Route 53), Firebase, Supabase, Docker, Kubernetes/Helm, Terraform
- **Observability & Practices:** Prometheus/Grafana, CloudWatch, PostHog, X-Ray; SLO/SLI tracking, distributed tracing; Git, Linux, microservices, agile, GDPR

Experience

Clover IT Services – New York, NY (Remote)

Jun 2022 – Jul 2024

Software Engineer / Site Reliability Engineer

- Designed, developed, and maintained **8 production microservices** across a multi-region AWS SaaS platform; owned features end-to-end from requirements through implementation, testing, and production deployment.
- Built backend services in **Python** (FastAPI, Flask) and **Node.js**, implementing RESTful APIs, data orchestration layers, and event-driven integrations (SQS/SNS) across 3 AWS regions.
- Developed reusable front-end components and internal tooling UIs, collaborating with product managers to translate requirements into well-architected user experiences.
- Established **pytest and Jest test gates** across 10+ services in CI/CD pipelines; reduced release lead time from 3 days to **4–6 hours (85% improvement)**.
- Architected full-stack observability (Prometheus/Grafana, CloudWatch, X-Ray) with SLO/SLI tracking; reduced **MTTR by 35%** through proactive alerting and root cause analysis.
- Designed self-service infrastructure using Terraform modules and containerised deployments (Docker, EKS/Helm); sustained **99.9% uptime** and delivered **22% infrastructure cost reduction**.
- **Promoted to team lead within 14 months**; mentored 3–4 engineers, led design reviews, and championed engineering best practices.

Seattle Kraken (NHL) – Seattle, WA

Jun 2021 – May 2022

Data Engineering Intern

- Built automated data pipelines in **Python** (PySpark, Azure Data Factory) for analytics ingestion and validation; reduced weekly report preparation time by **30%** across 500K+ records supporting real-time dashboards.
- Developed internal reporting tools and interactive dashboards using **React** and D3.js, enabling non-technical stakeholders to self-serve analytics without engineering involvement.
- Implemented data quality checks catching **15+** silent data corruption incidents before production; partnered with engineering to debug ETL reliability and improve data freshness.

Featured Projects

Purrrfect Keys – AI Piano Learning App

Jan 2026 – Present

expo.dev/accounts/ut254/projects/purrrfect-keys | *React Native, TypeScript, Firebase, Gemini AI, ONNX, Jest, Maestro*

- Shipped a cross-platform piano learning app through **14 development phases** with real-time pitch detection under **10ms**, AI-generated lessons via Gemini, competitive league with MMR ranking, and offline-first sync across 12 data types. **3,043 tests** passing across 149 suites.
- Designed the full architecture (React Native + Firebase + 9 Cloud Functions), built CI/CD with EAS Build, integrated PostHog analytics, and implemented gamification that drives daily active usage through cat avatar progression and streak rewards.

WhatsTheCraic – Live Event Discovery Platform

Apr 2025 – Jul 2025

whatsthecraic.run.place | *Node.js, React, FastAPI, MySQL, DynamoDB, Docker, AWS, Spotify OAuth*

- Launched a live event platform that improved event discovery conversion by **18%** through A/B-tested personalized ranking powered by Spotify listening data. Ingested events from Ticketmaster and Eventbrite with deduplication into a canonical data model.
- Designed and deployed **6 microservices** (aggregator, events, DJ, venue, auth, frontend) on AWS with Docker Compose, auto-TLS, and health/metrics instrumentation across 4 subdomains. Added ICS calendar export and Spotify OAuth for preference-based recommendations.

Genomic Benchmarking Platform – MSc Thesis

Sep 2025 – Dec 2025

github.com/UT07/qpAdm-Genomic-Benchmarking | *Terraform, Docker, Snakemake, EC2 Auto Scaling, S3, CloudWatch*

- Ran **10,700+ parallel genomic workloads** on AWS with zero manual intervention. Terraform provisioned Auto Scaling Groups that launched workers, executed containerised jobs, emitted CloudWatch metrics, and self-terminated automatically.
- Cut compute costs by **66%** using Spot Instances and eliminated **99%** of preprocessing overhead through data precomputation. Found the optimal instance configuration at **\$0.006 per run**.

Education

National College of Ireland, Dublin, Ireland

Sep 2024 – Jan 2026

MSc Cloud Computing

- **Coursework:** Cloud Architectures, Cloud DevOpsSec, Scalable Cloud Programming, Cloud Machine Learning, Data Governance/Compliance/Ethics, Research in Computing.

Southern New Hampshire University, Manchester, NH

Aug 2023 – May 2024

BS Computer Science

- **Coursework:** Software Dev Lifecycle, Full-Stack Development, Database Systems, Software Testing.

The University of Texas at Arlington, Arlington, TX

Aug 2019 – May 2022

BS Software Engineering (Coursework, Transferred)

- **Coursework:** Algorithms & DS, Operating Systems, Computer Networks, Information Security.
- **Teaching Assistant:** CSE 3318 Data Structures & Algorithms (50+ students).

Certifications

- *AWS Certified Solutions Architect – Professional (SAP-C02)*

Issued Mar 2024

- *AWS Certified Developer – Associate (DVA-C01)*

Issued Sep 2022

- *AWS Certified Cloud Practitioner (CLF-C01)*

Issued Jun 2022